Bugfix Log

Revision: 14 **Last modified:** 2026-06-13T16:00:00Z

Per CONST-MD-Bugfix-Documentation, every bug surfaced during implementation gets a permanent entry below: title, root cause, affected files, fix description, regression guard.

Entries are append-only; do not edit historical entries except to add clarification footnotes.

2026-04-27 â€” boba-jackett implementation (plan 2026-04-27-jackett-management-ui-and-system-db.md)

1. Master-key two-step write silent data-loss window

Severity: HIGH (silent credential loss possible on crash mid-bootstrap).

Root cause: Initial implementation of `bootstrap.EnsureMasterKey` generated `BOBA_MASTER_KEY` in memory, returned it to the caller, and *then* wrote `.env` in a separate step. If the process crashed, was killed, or the host lost power between those two operations, the caller would proceed to encrypt credentials with a key that disk had no record of â€” making them permanently undecipherable.

Affected files: - `qBitTorrent-go/internal/bootstrap/bootstrap.go` - `qBitTorrent-go/internal/envfile/write.go`

Fix: Collapse the generate + write into a single `envfile.Atomic` write (`tmpfile + fsync + rename`) before the function returns. Caller never sees a key that hasnâ€™t been durably persisted. Commit `c0f20bc`.

Regression guard: - `qBitTorrent-go/internal/bootstrap/bootstrap_test.go::TestEnsureMasterKeyDoesNotDuplicateHeader` - `qBitTorrent-go/internal/envfile/write_test.go` atomic-write test layer

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2. SQLite database file mode `0644` (world-readable credentials)

Severity: CRITICAL (any local user could read encrypted credentials + ciphertext-only attack surface).

Root cause: `modernc.org/sqlite` creates database files with the processâ€™s default umask, which on the container yielded `0644` (world-readable). The Go serviceâ€™s `.env` was already `0600` via `envfile.Atomic`, but `boba.db` was leaking â€” a defense-in-depth breach since the rows are encrypted, but still a violation of the principle â€œthe DB file should be readable only by the service ownerâ€.

Affected files: - `qBitTorrent-go/internal/db/conn.go`

Fix: After the first `Open()` succeeds, `db.Open` calls `os.Chmod(path, 0o600)` (and the `-wal` / `-shm` siblings if present), collapsing the file mode to owner-only. Commit `8405167`.

Regression guard: - Layer 4 security challenge `challenges/scripts/boba_db_file_perms_challenge.sh` - `qBitTorrent-go/internal/db/credential_leak_test.go`

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3. `AutoconfigResult` nil slices marshalled as JSON `null`

Severity: MEDIUM (silent client-breaking schema drift; UI rendered â€œundefinedâ€ badges instead of â€œ0 itemsâ€).

Root cause: Goâ€™s `json.Marshal` serializes a nil slice as `null`, not `[]`. When `Autoconfigure()` ran on a fresh DB with no discovered credentials, the `discovered_credentials`, `configured_now`, and `already_present` fields all marshalled as `null`. The OpenAPI 3.1 schema declared them as `array`, so clients (including the Angular dashboard) treated `null` as a contract violation.

Affected files: - `qBitTorrent-go/internal/jackett/autoconfig.go`

Fix: Pre-allocate empty slices (`[]string{}`) at the top of `Autoconfigure()` so a â€œnothing to doâ€ run still serializes as `{...,"discovered":[],"configured_now":[],...}`. Commit `6f3dbaf`.

Regression guard: - Layer 6 contract test `qBitTorrent-go/tests/contract/openapi_test.go` (validates each named field is `array` per OpenAPI schema even on empty runs).

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4. Catalog `ReplaceAll` empty-input wipe risk

Severity: HIGH (would permanently wipe the indexer catalog if Jackett returned an empty response; UI would show no indexers and fuzzy matcher would have nothing to match against).

Root cause: `repos.IndexerCatalog.ReplaceAll(rows []Row)` unconditionally `DELETE`d existing rows then `INSERT`ed the new ones. If `rows` was empty (Jackett momentarily unhealthy, transient 502, or catalog parse failure), the delete still happened, leaving an empty catalog table. Next refresh would have nothing to compare against.

Affected files: - `qBitTorrent-go/internal/db/catalog.go` (or equivalent repo file)

Fix: Refuse the operation early with `errors.New("repos: ReplaceAll refusing empty replacement")`. The caller (autoconfig orchestrator) treats this as a soft error, logs it, and leaves the previous catalog intact. Commit `8d71df1`.

Regression guard: - `qBitTorrent-go/internal/jackettapi/catalog_test.go::TestRefreshCatalogAllTemplatesFailReturns200WithErrors`

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5. `TestSearchHandler_QueueFull` pre-existing flake (NOT a boba-jackett bug)

Severity: LOW (test-only; flagged for follow-up).

Status: NOT FIXED in this plan. Confirmed unrelated to boba-jackett work.

Location: `qBitTorrent-go/internal/api/api_test.go`

Symptom: Occasional intermittent failure under load when the search queue fills before the testâ€™s `wait` returns.

Action: Logged here for traceability so future audits donâ€™t mis-attribute it. Open issue suggestion: stabilise by injecting a deterministic queue-full hook instead of racing against real goroutine scheduling.

2026-06-09 â€” Session 7 fixes (env_loader, AsyncMock, gamestorrents)

6. env_loader flaky test: KEY2 leak across test ordering

Severity: LOW (test-only flake; not a production bug).

Root cause: `test_comment_lines_ignored` in `tests/unit/test_env_loader.py` asserted `os.environ.get("KEY2") is None` after calling `load_env_files()` with a config file where `KEY2=commented_out` was commented out. However, `load_env_files` has a â€œfirst winsâ€ policy â€” if `KEY2` was already present in `os.environ` from a prior testâ€™s `load_env_files` call, the function would not override it. Under `pytest-randomly` ordering, the test that sets `KEY2` sometimes ran before the comment-line test, causing intermittent failure.

Affected files: - `tests/unit/test_env_loader.py`

Fix: Added explicit `os.environ.pop("KEY1", None)` and `os.environ.pop("KEY2", None)` at the START of the test (before calling `load_env_files`), ensuring a clean env state regardless of test execution order. The `finally` block was retained as a safety net.

Regression guard: - `tests/unit/test_env_loader.py::test_comment_lines_ignored` â€” ran 2147/2147 twice consecutively under random ordering with zero failures.

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7. AsyncMock warning in search deep-coverage tests

Severity: LOW (test warning; not a production bug).

Root cause: `test_iptorrents_login_no_cookies` in `tests/unit/merge_service/test_search_deep_coverage.py` used `AsyncMock()` for `login_resp` and `mock_session`. These objects are used as context managers (`async with session.post(...) as resp`), not as directly awaited coroutines. `AsyncMock.__aenter__` returns a coroutine, but the testâ€™s mock setup didnâ€™t properly wire `__aenter__`/`__aexit__`, causing â€œcoroutine was never awaitedâ€ RuntimeWarning.

Affected files: - `tests/unit/merge_service/test_search_deep_coverage.py`

Fix: Changed `AsyncMock()` to `MagicMock()` with explicit `__aenter__` and `__aexit__` stubs, matching the actual usage pattern (context manager, not coroutine).

Regression guard: - `tests/unit/merge_service/test_search_deep_coverage.py` â€” 0 AsyncMock warnings from this test (was 3).

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8. gamestorrents `_parse_size` B-substring bug (documented, not fixed)

Severity: MEDIUM (plugin returns 0 for all realistic file sizes).

Root cause: Same class of bug as BOB-013 (torrentkitty). The `multipliers` dict in `_parse_size` iterates in insertion order: `{"B": 1, "KB": 1024, ...}`. Since `"B"` is a substring of `"KB"`, `"MB"`, `"GB"`, and `"TB"`, the check `if unit in size_str` matches `"B"` first for all realistic sizes. Then `size_str.replace("B", "")` leaves a trailing unit character (e.g.Â `"35.2 G"`) which `float()` cannot parse, returning 0.

Affected files: - `plugins/gamestorrents.py` (line 74-86)

Fix: NOT FIXED in this session. Documented via tests that assert actual behavior (tests suffixed `_b_substring_bug`). Fix requires reordering the dict keys to check longest units first (same approach as BOB-013).

Regression guard: - `tests/unit/test_plugin_gamestorrents.py::TestParseSize` â€” 8 tests documenting the bug across GB/MB/KB/TB/commas/uppercase.

2026-06-09 â€” Session 8 parallel plugin tests + gamestorrents fix

9. gamestorrents `_parse_size` B-substring bug (fixed)

Severity: MEDIUM (plugin returned 0 for all realistic file sizes).

Root cause: Same class of bug as BOB-013 (torrentkitty). The `multipliers` dict in `_parse_size` iterated in insertion order: `{"B": 1, "KB": 1024, ...}`. Since `"B"` is a substring of `"KB"`, `"MB"`, `"GB"`, and `"TB"`, the check `if unit in size_str` matched `"B"` first for all realistic sizes. Then `size_str.replace("B", "")` left a trailing unit character (e.g.Â `"35.2 G"`) which `float()` could not parse, returning 0.

Affected files: - `plugins/gamestorrents.py` (line 74-86)

Fix: Reordered dict keys to check longest units first: `{"TB": ..., "GB": ..., "MB": ..., "KB": ..., "B": 1}`. Same approach as BOB-013.

Regression guard: - `tests/unit/test_plugin_gamestorrents.py::TestParseSize` â€” 8 tests all pass with correct byte values for GB/MB/KB/TB/B/comma/uppercase.

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10. piratebay `import os` placed after use (documented, not fixed)

Severity: LOW (only affects non-magnet torrent file downloads, which are rare).

Root cause: `piratebay.py:175` has `import os` placed after `os.fdopen` on line 172. When downloading a non-magnet torrent file, `os.fdopen` is called before `os` is imported, causing `UnboundLocalError`.

Affected files: - `plugins/piratebay.py` (line 172-175)

Fix: NOT FIXED in this session. Documented via test `test_torrent_file_download_known_bug_os_import_order`. Fix requires moving `import os` to the top of the `download_torrent` method or module level.

Regression guard: - `tests/unit/test_plugin_piratebay.py::TestDownloadTorrent::test_torrent_file_download_`

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11. torlock `search()` does not catch exceptions from `retrieve_url()`

Severity: LOW (network errors crash the entire search loop instead of failing gracefully).

Root cause: `torlock.py` `search()` calls `retrieve_url()` without a try/except, so a network error propagates up and crashes the search loop.

Affected files: - `plugins/torlock.py`

Fix: NOT FIXED in this session. Documented via test `test_search_exception_propagates`. Fix requires wrapping `retrieve_url()` in try/except with graceful error handling.

Regression guard: - `tests/unit/test_plugin_torlock.py::TestSearch::test_search_exception_propagates`

2026-06-09 â€” Session 8 wave 2: nyaa/kickass/anilibra/torrentgalaxy/yts

12. nyaa `download_torrent` missing `import re` (documented, not fixed)

Severity: MEDIUM (any nyaa.si URL passed to `download_torrent` raises `NameError`).

Root cause: `plugins/nyaa.py:170` calls `re.search()` but `import re` is absent from the module. The `search()` method works because `re` is imported transitively via other modules, but `download_torrent` fails with `NameError: name 're' is not defined`.

Affected files: - `plugins/nyaa.py` (line 170)

Fix: NOT FIXED in this session. Documented via tests that assert the `NameError`. Fix requires adding `import re` at the top of the file.

Regression guard: - `tests/unit/test_plugin_nyaa.py::TestDownloadTorrent` â€” 6 tests including the NameError documentation.

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13. kickass comma-separated size not matched by regex (documented, not fixed)

Severity: LOW (sizes like `1,234.5 MB` silently parse to 0).

Root cause: The HTMLParser regex `\d+\.\d+` doesnâ€™t match comma-separated numbers like `1,234.5 MB`. The comma is not handled in the regex or the replacement logic.

Affected files: - `plugins/kickass.py`

Fix: NOT FIXED in this session. Documented via test `test_comma_in_size_not_matched_by_regex`. Fix requires updating the regex to handle commas.

Regression guard: - `tests/unit/test_plugin_kickass.py::TestHTMLParserFeed::test_comma_in_size_not_matched`

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14. kickass crash-prone patterns (fixed)

Severity: MEDIUM (empty responses from `retrieve_url` crash the plugin).

Root cause: Three methods in `kickass.py` called `retrieve_url()` without try/except or empty-response guards: `__retrieve_download_link()`, `download_torrent()`, and `search()`. When `retrieve_url` returned `None` or empty string, subsequent `re.search`/`re.sub` calls crashed with `TypeError`.

Affected files: - `plugins/kickass.py` (lines 61, 74, 94)

Fix: Added try/except + empty-response guards to all three methods. Each now handles empty/None responses gracefully (returns â€œNotFoundâ€, prints fallback URL, or breaks the loop respectively).

Regression guard: - `tests/unit/test_plugin_kickass_guards.py` â€” 13 tests proving empty response, None response, ConnectionError, TimeoutError, and malformed HTML donâ€™t crash.

2026-06-09 â€” Waves 5-8: B-substring epidemic across 9 community plugins

15. Systemic `_parse_size` B-substring bug (9 instances found, all fixed)

Severity: MEDIUM (all 9 plugins returned 0 for KB/MB/GB/TB â€” silently broken size reporting).

Root cause: The `_parse_size` method in each plugin uses a `multipliers` dict iterated in insertion order. With keys ordered `{"B": 1, "KB": 1024, ...}`, the check `"B" in size_str` matches the `"B"` substring inside `"KB"`, `"MB"`, `"GB"`, and `"TB"`. Then `size_str.replace("B", "")` leaves a trailing character (e.g.Â `"1.5 G"`) which `float()` cannot parse, returning 0.

Affected plugins (9 total): | Plugin | Status | |â€”â€“â€“|â€”â€“â€“| | `gamestorrents.py` | Fixed (BOB-024) | | `megapeer.py` | Fixed (BOB-042) | | `one337x.py` | Fixed (BOB-043) | | `extratorrent.py` | Fixed (BOB-044) | | `torrentfunk.py` | Fixed (BOB-045) | | `therarbg.py` | Fixed (BOB-047) | | `pctorrent.py` | Fixed (BOB-052, by subagent) | | `bitru.py` | Fixed (BOB-054) | | `xfsub.py` | Fixed (BOB-057) | | `yihua.py` | Fixed (BOB-058) |

Fix: Reorder dict keys longest-unit-first: `{"TB": ..., "GB": ..., "MB": ..., "KB": ..., "B": 1}`. Same approach as BOB-013 (torrentkitty).

Global regression guard: 10 dedicated test files, each with explicit `_parse_size` assertions for all 5 units (B/KB/MB/GB/TB).

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16. Systematic `import re` omissions (3 instances found, all fixed)

Root cause: Three plugins called `re.search()` or `re.compile()` without importing `re` at module level. The `search()` method worked via transitive imports from helper modules, but `download_torrent()` failed with `NameError`.

Affected plugins: | Plugin | Status | |â€”â€“â€“|â€”â€“â€“| | `nyaa.py` | Fixed (BOB-035) | | `audiobookbay.py` | Fixed (BOB-042) | | (torlock `search()` also lacks exception handling â€” documented BOB-029) |

Fix: Added `import re` at module level.

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17. bt4g infinite loop (test-only, fixed)

Severity: LOW (test hang, not production).

Root cause: `search()` has a `while True` loop that breaks when `parser.noTorrents` is True. Tests that used `return_value=MATCHING_HTML` (not `side_effect`) caused every page to match, triggering infinite loop.

Fix: Changed to `side_effect=[MATCHING_HTML, EMPTY_HTML]` so page 2 triggers the `noTorrents` break condition.

2026-06-13 â€” BobaLink extension: flaky-perf-test hardening + WCAG contrast fixes

Batch verified together by `extension/ci-ext.sh` â†’ `CI-EXT: PASS`, full suite **632 passed (632)**, `tsc --noEmit` clean, `npm run lint` 0/0, chrome+firefox builds loadable, Â§11.4.38 asset-verify pass, both store zips â‰¥10 KiB. Every fix below carries a permanent regression guard (Â§11.4.135) that was RED before the fix.

18. Junk-flood DoS security test â€” load-coupled absolute-wall-clock FAIL-bluff

Severity: MEDIUM (flaky test â€” Â§11.4.1 FAIL-bluff; blocked the green gate).

Root cause (FACT, not guess): `extension/tests/security/scanner-hostile-input.test.ts` asserted a tight absolute `expect(elapsed).toBeLessThan(5000)` on a 50k-anchor orchestrator scan. On a shared/oversubscribed host (load-avg 15) the *correct* scan measured **6608 ms**; in isolation it passes <5000 ms (proven 5/5). Product behaviour was always correct (exactly 2 magnets detected, all junk excluded) â€" only the absolute timing tripped.

Affected files: `extension/tests/security/scanner-hostile-input.test.ts`.

Fix: replaced the tight 5000 ms budget with a generous **30 s hang-ceiling** (catches a true hang / quadratic explosion â€" a real O(nÂ²) over 50k anchors takes minutes) and moved the rigorous machine-independent DoS guard to a new perf test. Correctness assertions retained.

Regression guard: new `extension/tests/perf/orchestrator-scaling.perf.test.ts` asserts a metamorphic sub-quadratic scaling ratio (`t(10Â·N)/t(N) < 30`; linearâ‰ˆ10, quadraticâ‰ˆ100 â€" dimensionless, immune to host load) with a golden-good/golden-bad oracle self-validation (Â§11.4.107(10)): `linearRatio=9.9` accepted, `quadraticRatio=113` rejected.

19. crypto.perf p99 budget â€" contention-coupled FAIL-bluff

Severity: MEDIUM (flaky test).

Root cause (FACT): `extension/tests/perf/crypto.perf.test.ts` asserted the 80 ms budget on `dist.p99`. PBKDF2 Ã—2 is CPU-bound; under concurrent-suite contention p99 spiked to **164 ms** while intrinsic cost stayed ~23â€"41 ms (proven in isolation). The newly-added heavy parallel perf/stress suites added the contention that exposed it.

Affected files: `extension/tests/perf/crypto.perf.test.ts`.

Fix: assert the budget on `dist.min` (intrinsic, contention-robust â€" contention only adds time). A real â‰¥3.6× algorithmic regression still inflates the min past 80 ms; correctness (EXACT plaintext recovery) is asserted every round.

Regression guard: the same test (now contention-robust); isolated min ~23 ms â‰¤ 80; full-suite `CI-EXT: PASS`.

20. parsers.perf scaling-ratio â€" contention-coupled FAIL-bluff

Severity: MEDIUM (flaky test).

Root cause (FACT): `extension/tests/perf/parsers.perf.test.ts` computed the per-link scaling ratio from the **median** time at each size. The small (1000-link, ~5 ms) median is noise-sensitive under contention; the ratio tripped **3.09 > 3** while the intrinsic ratio is ~1.0 (proven 1.34 / 1.00 in isolation).

Affected files: `extension/tests/perf/parsers.perf.test.ts`.

Fix: compute per-link cost from the **min** run at each size (intrinsic). A genuine O(nÂ²) regression still shows ratio â‰ˆ5 in the min run, so the cap of 3 keeps full regression-catching power while never flaking.

Regression guard: the same test; isolated ratio 1.24 â‰¤ 3; full-suite `CI-EXT: PASS`.

21. Three real WCAG 2.1 AA colour-contrast defects (popup theme tokens)

Severity: MEDIUM (accessibility â€” WCAG SC 1.4.3, real product defect).

Root cause (computed, Â§11.4.6): three popup theme tokens fell below the 4.5:1 normal-text contrast floor. Found by the new `extension/tests/a11y/focus-and-contrast.a11y.test.ts` (computes contrast ratios from the REAL committed CSS; analyzer self-validated against WCAG reference extremes), kept RED until fixed.

Affected files: `extension/src/popup/styles.css` (each token in both the `@media prefers-color-scheme` block and the explicit `[data-theme]` class).

Fix (same hue family, minimal visual change, â‰¥4.6:1): - `--text-faint` dark on `#1a1a2e`: `#7a7a90` (4.07) â†' `#838399` (**4.61**) - `--text-faint` light on `#f5f6fa`: `#8a8a9c` (3.14) â†' `#6c6c7e` (**4.76**) - `--warning-text` light on `#fff6e0`: `#9a6a00` (4.40) â†' `#946400` (**4.78**)

Regression guard: the 3 a11y tests flipped REDâ†'GREEN on the fix (proving they catch the defect) and are permanent Â§11.4.135 guards.

2026-06-13 â€" BobaLink extension wave-11: options-page WCAG a11y fixes

Found by the new `extension/tests/a11y/options-contrast-motion.a11y.test.ts` (computes contrast ratios from the REAL committed `src/options/styles.css`; analyzer self-validated against WCAG reference extremes), kept RED until fixed. Batch verified by `extension/ci-ext.sh` â†' `CI-EXT: PASS` with the wave-11 coverage additions. All REDâ†'GREEN proven; the Â§11.4.120 reconciliation rewrote the gates to assert the NEW mechanism (not fake-passed, not reverted).

22. Save-button text below WCAG AA over its gradient (SC 1.4.3)

Severity: MEDIUM (accessibility, real). **Root cause (computed):** `.btn-primary` has `color:#fff` over `linear-gradient(--accent â†' --accent-2)`; white on the lighter `--accent` (#667eea) end is only **3.66:1** (14px normal-weight text needs â‰¥4.5). **Fix:** the global `--accent` is ALSO `.nav-item.active` text on a dark sidebar where darkening it would REDUCE contrast, so the button uses a **button-local** darker indigo start `#5a6fce` (**4.57:1**); the `--accent-2` end is already 6.37:1. **Affected:** `src/options/styles.css` `.btn-primary`. **Guard:** the test now reads the buttonâ€™s ACTUAL gradient stops and asserts #fff â‰¥ 4.5 over every stop.

23 & 24. Field labels + nav-hover invisible in light theme â€" hard-coded literals (SC 1.4.3)

Severity: MEDIUM (accessibility, real â€" labels effectively unreadable in light mode). **Root cause (computed):** `.field > label` (`#d0d0e0`) and `.nav-item:hover` (`#e0e0e0`) were hard-coded **dark-theme** literals that never re-resolved per theme â†' **1.40:1** and **1.12:1** on the light backgrounds. **Fix:** tokenized to theme-aware tokens â€" label â†' `var(--text)` (10.6/13:1), nav-hover â†' `var(--text-strong)` (15.5/16.1:1) â€" legible in both themes. **Affected:** `src/options/styles.css`. **Guard:** the Â§11.4.120-reconciled test asserts the

selectors now use a `var(--token)` (not a literal) AND the resolved token clears AA in BOTH themes.

25 & 26. No prefers-reduced-motion escape (SC 2.3.3 / 2.2.2)

Severity: MEDIUM (accessibility, real). **Root cause:** the options CSS ships `@keyframes fadeIn` (panel reveal on tab activation) + transitions, and the popup CSS ships transitions, but neither had a `@media (prefers-reduced-motion: reduce)` block â€" motion-sensitive users had no way to suppress it. **Fix:** added the standard reduced-motion block (zeroing animation/transition durations) to BOTH `src/options/styles.css` and `src/popup/styles.css`. **Guard:** the a11y tests assert each stylesheet ships non-trivial motion AND a reduced-motion block (REDâ†'GREEN on the fix).

2026-06-13 â€" BobaLink extension wave-12: scaling-ratio FAIL-bluff hardening

Batch verified by `extension/ci-ext.sh` â†' `CI-EXT: PASS`, full suite **799 passed (799)** (+38 over wave-11: crypto-tamper 17, link-scanner-coverage 10, highlight-manager 11 â€" all green, no product defects). The one fix below was a flaky test exposed by the heavier 799-test load.

27. infohash dedup scaling-ratio test â€" single-run + tiny-floor FAIL-bluff (Â§11.4.50)

Severity: MEDIUM (flaky test â€" Â§11.4.1 FAIL-bluff; blocked the green gate). **Root cause (FACT):** `tests/security/infohash-detection-hostile.test.ts`â€™s relative-scaling guard used a SINGLE-run `tLarge / Math.max(tSmall, 0.05)`; on a sub-ms baseline the 0.05 ms floor made the ratio noise-dominated (**53.6** observed under full-suite contention vs ~4 intrinsic). Worse, the â‰¤40 threshold was too loose to even catch the quadratic it guards (4Ã— input â†' linearâ‰ˆ4, quadraticâ‰ˆ16 â€" both <40). Passed at 761 tests, tripped at 799. **Fix:** measure the MIN over 7 reps at each size (intrinsic, contention-robust â€" host stalls only ADD time) and tighten the threshold to **10** (sits BETWEEN linearâ‰ˆ4 and quadraticâ‰ˆ16, so it now genuinely catches an O(nÂ²) regression while never flaking). Verified stable 5/5 in isolation + full-suite `CI-EXT: PASS`. **Affected:** `extension/tests/security/infohash-detection-hostile.test.ts`. **Note:** the sibling stress testâ€™s median estimator was initially assessed robust â€" but it ALSO flaked one wave later under heavier load (see entry 28), confirming that median is not enough; min is.

2026-06-13 â€" BobaLink extension wave-13: tab-group scaling medianâ†'min + working-tree hygiene

Batch verified by `extension/ci-ext.sh` â†' `CI-EXT: PASS`, full suite **814 passed (814)** (+15 over wave-12: `popup-states` 5 â€" genuine partial-Send-All gap; plus two prior-session test files brought into git â€" `options-save-flow` 4 + `offline-queue-persistence` 6 â€" that were sitting UNTRACKED in the working tree). No product defects.

28. tab-group scaling-ratio test â€” median estimator spikes under load (Â§11.4.50)

Severity: MEDIUM (flaky test â€” Â§11.4.1 FAIL-bluff; blocked the green gate). **Root cause (FACT):** `tests/stress/orchestrator-ratelimiter-tabgroup.stress.test.ts`â€™s scaling guard used the MEDIAN of 5 reps; under the heaviest concurrent load (814 tests) several reps landed on a busy core and dragged the median up, so the 10Ã—-input ratio spiked past the 25 ceiling while the intrinsic ratio is ~5â€“8. (This refutes the wave-12 assessment that median was robust â€” entry 27â€™s note.) **Fix:** switched the shared timing helper from median to **MIN** over reps (the minimum is the intrinsic-cost estimator â€” host stalls only ADD time, so the fastest run is closest to true cost; a genuine O(nÂ²) regression still inflates every run including the min) and bumped reps (orchestrator 3â†’5, tab-group 5â†’9). Verified stable 3/3 in isolation (ratios 4.76/6.41/8.25 vs the 25 ceiling) + full-suite `CI-EXT: PASS`. **Affected:** `extension/tests/stress/orchestrator-ratelimiter-tabgroup.stress.test.ts`. **Lesson (Â§11.4.118):** the robust estimator for CPU-bound scaling-ratio tests under concurrent load is MIN-of-reps, not median/mean â€” all six flaky-scaling fixes this session converge on it.

2026-06-13 â€” BobaLink extension wave-14: live-7187 harness hardening

Live-harness prep to de-risk the operator-gated live round-trip. The default suite is unchanged at **814 passed (814)** (`CI-EXT: PASS`); the live test + challenge are out-of-suite and operator-run.

29. live_detect_send challenge â€” Â§11.4.14 cleanup-race leaves an orphan synthetic torrent

Severity: MEDIUM (anti-bluff/hygiene defect in a live Challenge). **Root cause (FACT):** `challenges/extension/live_detect_send_challenge.sh` cleaned up the synthetic torrent it adds ONLY when its secondary qBittorrent `torrents/info` probe reported `present`. But the proxy adds the magnet itself; that separate probe can race/auth-fail and read `absent` while the magnet IS added â€” so a PASS could leave an orphan synthetic torrent in qBittorrent (a Â§11.4.14 quiescence violation). **Fix:** cleanup now keys off the **proxyâ€™s authoritative per-url verdict** (`results[0].status == "added"`) OR the probe, re-logs-in for a fresh SID when needed, and warns honestly if delete-login fails (never reds the run at teardown). Also fixed a `/bin/sh -n` parse issue (apostrophe/metachar in added comments â€” Â§11.4.67). **Affected:** `challenges/extension/live_detect_send_challenge.sh`. **Verification:** clean SKIP (exit 77) when the backend is down (no fail-open), `bash -n` + `sh -n` PASS, mutation-verified (no-op/partial response stubs â†’ FAIL); the cleanup trigger confirmed to fire on a simulated proxy-`added` + probe-`absent` response (the old code would not). The up-path runs when the operator brings the stack up.

2026-06-13 â€” Boba broader-project (merge-service / Go / plugins / frontend) verification + 2 fixes

A full-project verification pass (parallel subagents, all run against the REAL toolchains, anti-bluff captured evidence) confirmed the broader Boba backend is green: **Go** `qBitTorrent-go` **14 packages PASS under** `-race`**, vet clean, no data races**; **plugins: 60** `.py` **compile, 12 curated**

contract-valid, 901 plugin tests pass (under `.venv`/py3.13); **Angular frontend: build GREEN + 342 unit tests pass**; **merge-service Python: 4149 unit tests pass, ruff clean** (under `.venv`/py3.13). Two genuine items found:

30. `ci.sh` / `scripts/run-tests.sh` false-red under a host `python3` < 3.12

Severity: MEDIUM (the projectâ€™s own manual CI gate false-reds â€” a Â§11.4.1 FAIL-bluff at the toolchain layer). **Root cause (FACT, independently found by two subagents):** both scripts invoke a **bare** `python3`, which resolves to the host default. `pyproject.toml` declares `requires-python = ">=3.12"` and the suite uses `tomllib` + PEP-604 `X | Y` union syntax that a `python3` < 3.12 cannot even COLLECT. On this host `python3` = **3.9.25**, so `python3 -m pytest tests/unit/` aborts: **â€œInterrupted: 36 errors during collectionâ€** (`TypeError: unsupported operand`), while the project `.venv` (py3.13) collects **4149** and passes all. **Fix:** both scripts now select a `>=3.12` interpreter via a `_select_python` helper (prefers `$PYTHON` â†’ `.venv/bin/python` â†’ `python3.13` â†’ `python3.12` â†’ a bare `python3` only if it is â‰¥3.12), aborting with a clear message if none qualifies. Backward-compatible: a host already on `python3` â‰¥ `3.12` selects the same interpreter. **Affected:** `ci.sh` (lines for py_compile + the unit/integration/e2e pytest steps), `scripts/run-tests.sh` (the 5 `exec â€¦ pytest` modes). **Verification (captured):** `bash -n` clean both; the selector picks `.venv` py3.13; bare py3.9 â†’ â€œ36 errors during collectionâ€ vs selected interpreter â†’ â€œ4149 tests collectedâ€ on the same `tests/unit/` dir.

31. Go backend â€” 2 uncovered critical paths closed (coverage, not a defect)

Type: Task (coverage). The `qBitTorrent-go` `merge_search` `RunSearch` goroutine-orchestration loop (every prior test passed a `nil` client) and the `sse_broker` concurrent pub/sub had zero real coverage. Added 4 anti-bluff tests (`internal/service/coverage_test.go`, real `httptest` qBittorrent server, no mocks) â€” RunSearch completes/accumulates + StartFailsâ†’failed + ContextCancelâ†’abort, and an 8-publisher/16-subscriber SSEBroker churn under `-race`. **Anti-bluff proof:** a no-op `return nil` stub in `RunSearch` made all three RunSearch tests FAIL, then reverted (production clean, Â§11.4.84). Verified GREEN under `-race` (service package `ok 5.933s`, no data race). Production code unchanged.

2026-06-13 â€” Boba merge-service hermetic specialized suites: 5 test-quality fixes (product correct)

A verification pass over the merge-serviceâ€™s hermetic specialized suites (`stress, chaos, property, contract, concurrency, memory, observability, security` â€” run under `.venv`/py3.13, the suites BG-1â€™s `tests/unit` pass did not cover) found the PRODUCT is correct + secure (bandit 0 HIGH; CORS hardened; no product defect) but surfaced 5 real TEST-quality defects across the anti-bluff classes. All test-side only â€” `download-proxy/src/` unchanged; each verified green under `.venv`.

32. Five merge-service test-quality defects (Â§11.4.1 / Â§11.4.3 / Â§11.4.50 / Â§11.4.120)

- `tests/stress/test_pipeline_stress.py` **â€” Â§11.4.50 flaky absolute-wall-clock threshold.** Gated on `max(latencies_ms) < 2000` (absolute); a lone GC/scheduler stall spiked one iteration to ~140 ms (~25Ã— the 5.6 ms median) â€” an absolute bound is the wrong oracle. Fixed to a RELATIVE bound `max â‰¤ 200 Ã— max(p50, 1.0 ms)` (an O(nÂ²) blowup inflates the median too, so it still fires; a lone outlier no longer flakes). Verified: 3 passed.
- `tests/contract/test_crossapp_theme_contract.py` **â€” Â§11.4.1/ Â§11.4.3 FAIL-instead-of-SKIP.** The 3 `@requires_compose` testsâ€™ own docstring promises â€œskip cleanly when the proxy is unreachableâ€, but `_fetch()` did a bare `urlopen` â†’ `URLError` â†’ FAIL when down. Fixed `_fetch()` to catch only CONNECTION errors â†’ `pytest.skip` (with `./start.sh -p` reason); `HTTPError` still returns the real status and `AssertionError` is NOT swallowed (a real missing-bridge still FAILs when the proxy is up). Verified: 3 SKIP when down, suite 27 passed when collectable.
- `tests/security/test_cors.py` **â€” Â§11.4.120 stale gate asserting REMOVED insecure behavior.** `TestCORSWildcardDefault` asserted `Access-Control-Allow-Origin == "*"` as the default, but the product was deliberately hardened to a secure-by-default localhost/dev allowlist (wildcard removed, CONTINUATION known-issue #5). The TEST lagged the security improvement. Reconciled to assert the NEW secure default â€” a known localhost origin reflected EXACTLY, an unknown LAN origin NOT reflected and never `*`. Anti-bluff: a wildcard-revert fails the `!= "*"` assertions (RED-on-old/GREEN-on-new), not a tautology. Verified.
- `tests/observability/test_metrics_exist.py` **â€” Â§11.4.3 infra-FAIL instead of SKIP.** A `@requires_compose` test made an unconditional `httpx.get(:7187/metrics)` â†’ `ConnectError` FAIL when down. Added the existing `@merge_service_required` skip guard â†’ SKIPs cleanly. Verified.
- `tests/security/test_jackett_autoconfig_secrets.py` **â€” toolchain-path FAIL.** `subprocess.run(["bandit"])` raised `FileNotFoundError` (bandit is in `.venv/bin`, not on bare PATH). Added a `_resolve_bandit()` helper (prefers the running interpreterâ€™s bin dir, then `shutil.which`, SKIP if genuinely absent). Product clean: `.venv/bin/bandit` on `jackett_autoconfig.py` â†’ 0 HIGH findings. Verified.

2026-06-13 â€” Boba broken-doc-links + perf-test container-boot (the last test-type suites)

Ran the remaining test types (performance/benchmark/load/docs) under `.venv`. Benchmark: 9 infra-free pass (500Ã—â€“4760Ã— threshold headroom â€” not flaky, correctly left unchanged) + 5 by-design error-not-skip infra-down (intentional). Load: only a Locust `locustfile.py` (no pytest tests). Two real items fixed (BUGFIXES 33, 34).

33. `docs/` test caught 8 broken internal markdown links + 1 test FAIL-bluff

Severity: LOW-MEDIUM (broken doc links + a Â§11.4.1 test FAIL-bluff). **Root cause + fix (FACT):** `tests/docs/test_no_broken_links.py` (resolves links relative to each docâ€™s own dir, the markdown convention) flagged 8 docs. Seven were genuine broken links â€” `docs/*_Summary.md` linked to siblings with a spurious `docs/` prefix (`[Architecture Summary](docs/Architecture_Summary.md)` in a doc already

inside `docs/` â†' resolves to `docs/docs/...`). Fixed by dropping the prefix (`](docs/X)` â†' `](X)`); `docs/Scripts_Summary.md`â€™s `](scripts/pre_build_verification.sh)` â†' `](../scripts/...)` (the script is at repo-root). The 8th â€" a research doc â€" was a **TEST FALL-BLUFF**: the link extractor matched `[${tab.title}](${tab.url})` inside a **JS code block** (not a doc link). Fixed the extractor to STRIP fenced + inline code before matching. Anti-bluff verified (Â§1.1): a real broken link in **prose** still FAILs (the strip only removes code, not real links â€" not a tautology). **Affected:** 7 `docs/*_Summary.md` (+regenerated `.html`/`.pdf`), `tests/docs/test_no_broken_links.py`. **Verified:** `tests/docs` â†' 129 passed / 0 failed / 2 honest skip (mkdocs absent).

34. Performance test BOOTED CONTAINERS + hung 120 s on infra-down (Â§11.4.3 / no-boot)

Severity: MEDIUM (a test that actively boots infra + hangs). **Root cause (FACT):** `tests/performance/test_concurrent_search.py::TestConcurrentSearch` consumed a `merge_service_live` fixture that, on ports-down, ran `podman/docker compose up -d` then `_wait_for_port(timeout=120)` â€" pytest-timeout killed the wait â†' 7 setup-ERRORs + 2 min wall-clock, and it was ACTIVELY booting containers. The suiteâ€™s siblings (`stress`/`security`/`observability`) all use the fast `merge_service_required` probe-and-SKIP guard. **Fix:** applied that same `merge_service_required` guard (1 s socket+`/health` probe, never boots containers) â†' 7 honest infra-down SKIPs. Non-tautology: when the stack is up the class still runs every real assertion (status 200, latency ceilings, p50/p95) unchanged. (The benchmark suiteâ€™s separate error-not-skip design is INTENTIONAL and was left untouched per Â§11.4.122/Â§11.4.6.) **Affected:** `tests/performance/test_concurrent_search.py`. **Verified:** 7 SKIP cleanly, no container boot, `download-proxy/src/` unchanged.

2026-06-13 â€" `start.sh` boot completely broken: `boba-ctl up -d` flag mismatch

35. `./start.sh` aborted EVERY boot at the container-start step (`-d` flag)

Severity: HIGH (the documented startup path `./start.sh` did not work at all). **Root cause (FACT, found by actually running the boot):** `start.sh::start_container()` runs the generic `$COMPOSE_CMD up -d`. In the default boba-ctl mode `$COMPOSE_CMD` = `scripts/boba-ctl.sh`, whose `up|down)` case did `exec "$BOBA_CTL_BIN" "$@"` â€" passing the compose-style `-d` (detach) straight to the `cmd/boba-ctl/boba-ctl` Go binary, whose `up` defines only `-profile`/`-wait`. Result every boot: `flag provided but not defined: -d` â†' `[ERROR] Failed to start container` â†' `exit 1`, before any container was created. **Fix:** `scripts/boba-ctl.sh`â€™s `up|down)` case now strips `-d`/`--detach` (boba-ctl runs detached by default) so it is a true drop-in for `compose up -d`. **Affected:** `scripts/boba-ctl.sh`. **Verified:** `bash -n` clean; the re-run boot proceeded past the step into `podman-compose ... up -d` (created `pod_boba`, pulled all images).

Operator-environment note (NOT a project defect, reported per Â§11.4.6): with the `-d` bug fixed, the boot then HUNG inside `podman-compose` itself â€" it created `pod_boba` + pulled every image, then froze at 0.0% CPU (`STAT=SN`) before creating any container. `podman` itself works (`podman run alpine` â†' OK), so the hang is specific to `podman-compose`, which on this host runs under homebrew `python@3.14` (a likely-incompatible combo; podman-compose support for 3.14 is not established). Native `podman compose` just delegates to the

same `podman-compose`. This is an operator-environment toolchain issue (fix: run podman-compose under a supported python, e.g.Â 3.11/3.12, or repair the podman-compose install), surfaced to the operator â€" the projectâ€™s `./start.sh` orchestrator code is now correct.

Clarifying footnote (2026-06-13, root-caused â€" supersedes the python@3.14 hypothesis above): the hang was NOT python@3.14. Captured live: the `vfkit` VM shares are `/Users`, `/private`, `/var/folders` and `podman machine inspect â†' Mounts:[]` â€" the repo lives on `/Volumes/T7` (external SSD), which is NOT shared into the podman applehv VM, so the first `podman create` stalls on an unreachable bind path. Proven by both python 3.13 AND 3.14 hanging identically (refuting the version theory). Fix (operator-authorised, Â§11.4.122): recreated the machine with `--volume /Volumes/T7:/Volumes/T7 --cpus 4 --memory 6144`; all 5 containers then booted healthy. On macOS the `network_mode: host` ports live in the VM, so an SSH `-L` tunnel (or `scripts/ensure-macos-tunnel.sh`) bridges :7186/:7187 to the Mac.

2026-06-13 â€" qBittorrent 5.x API-compat: download round-trip broken (login + add)

Surfaced by the live detectâ†'sendâ†'torrent round-trip (`extension/tests/live/download-endpoint.live.test.ts`) once the stack was finally booted (qBittorrent **v5.2.1**, `linuxserver/qbittorrent:latest`). Three real PRODUCT defects + two test/challenge-quality items, each proven with captured live evidence. The whole class: the proxy + tests spoke legacy qBittorrentâ€™s plain-text protocol (`200 "Ok."`) while the image speaks 5.xâ€™s JSON/204.

36. Proxy reported `auth_failed` for EVERY download â€" modern qBittorrent 204 login

Severity: CRITICAL (no download could reach qBittorrent). **Type:** Bug. **Root cause (FACT, captured live):** `download-proxy/src/api/routes.py` judged WebUI login with `resp.status == 200 and body == "Ok."`. Modern qBittorrent (4.6+/5.x) returns `204 No Content` with an EMPTY body + the `QBT_SID` cookie (`HTTP/1.1 204` / `set-cookie: QBT_SID_7185=â€¦`) â†' `status != 200` AND `body != "Ok."` â†' `{"status":"auth_failed"}` even though login succeeded. The defect lived at FOUR login call-sites (`/download`, `/download/upload`, `/downloads/active`, `/auth/qbittorrent`) â€" the active-downloads dashboard widget + credential-save endpoint were broken too. **Fix:** new `_qbit_login_succeeded(status, body, cookies)` â€" authoritative version-independent signal is â€œthe server issued a `QBT_SID`/`QBT_SID_<port>` session cookieâ€ (exact/prefix match, not a loose `"SID"` substring), legacy `Ok.` body as fallback; all four call-sites migrated. **Affected:** `download-proxy/src/api/routes.py`. **Regression guard (Â§11.4.135):** `tests/unit/test_qbit_login_compat.py` incl.Â `test_reproduces_pre_fix_defect_old_check_rejected_modern_204` (Â§11.4.115 polarity). **Verified live:** `POST /api/v1/auth/qbittorrent admin/admin` â†' `{"status":"authenticated","version":"v5.2.1"}`; the round-trip went from auth_failed-SKIP to GREEN.

37. Proxy reported `failed` for successful adds â€" modern qBittorrent JSON add + duplicate 409

Severity: CRITICAL (a torrent that actually landed was reported failed). **Type:** Bug. **Root cause (FACT, captured live):** the add-success check was `body.lower().startswith("ok")`.

Modern qBittorrent `/api/v2/torrents/add` returns a JSON summary, not `Ok.`: `{"added_torrent_ids": ["<hash>"],"failure_count":0,"pending_count":0,"success_count":1}` â€" the torrent IS added (confirmed in `/torrents/info`) but the proxy saw `{` â†' `failed` with the JSON as `detail`. SECOND, related: BobaClient retries on network/timeout; attempt 1 added the magnet but was slow â†' client timed out â†' retry re-sent the SAME infohash â†' qBittorrent `409 Conflict` (duplicate) â†' `failed` (proven: one round-trip = TWO `/api/v1/download` requests in the proxy log). **Fix:** new `_qbit_add_succeeded(status, body)` â€" accepts the modern JSON (`added_torrent_ids` non-empty OR `success_count`/`pending_count` >= 1, with defensive int-coercion so a malformed `"N/A"` count classifies as failure, never crashes the add path), legacy `Ok.`, AND treats `409 Conflict` as idempotent SUCCESS (the torrent IS present â€" makes a retried add safe); all three add call-sites migrated. **Affected:** `download-proxy/src/api/routes.py`. **Regression guard:** `tests/unit/test_qbit_login_compat.py` (modern-JSON / pending-only / all-failed / 409-duplicate / non-numeric-count / legacy-Ok / Fails / 415 cases). **Verified live (Ã—3, Â§11.4.50):** the live round-trip + `challenges/extension/live_detect_send_challenge.sh` both GREEN â€" the synthetic infohash is INDEPENDENTLY confirmed present in qBittorrentâ€™s real `/torrents/info`, then cleaned up (Â§11.4.14).

38. Live test + challenge could not cross-check modern qBittorrent (test-quality, no product defect)

Severity: MEDIUM (anti-bluff held: tests SKIPâ€™d honestly, never false-PASSâ€™d). **Type:** Task. **Root cause:** (a) the live testâ€™s own confirm-login helper had the same legacy `status==200 && "Ok."` assumption â†' could not read `/torrents/info` on 204 â†' honest SKIP. (b) the challenge cross-checked qBittorrent at `:7185` (container-internal, not reachable off-host on macOS). **Fix:** (a) confirm-login now accepts 200/204 + keys on the `Set-Cookie` SID (mirrors the proxy); (b) challenge default qBittorrent base `:7185` â†' `:7186` (the documented WebUI access port, reachable on Linux + macOS). **Affected:** `extension/tests/live/download-endpoint.live.test.ts`, `challenges/extension/live_detect_send_challenge.sh`. **Verified:** both GREEN with real infohash-present confirmation + Â§11.4.14 cleanup.

Code review (Â§11.4.142/Â§11.4.134): an independent adversarial review caught 2 BLOCKING (2 of the 4 login sites initially missed; an `int("N/A")` crash path) + 1 warning (loose `"SID"` substring) â€" all remediated + guarded; the re-review returned a clean GO.